AI TRAINING – ASSIGNMENT 1

Data Visualization with Matplotlib

Line, bar, histogram, scatter, labels and legends

1. Objective

Complete the practical tasks using Python. Run your code, check the output, and answer the short questions in your own words.

2. Assignment Tasks

Part A – Line and Bar Charts

1. Create a line chart showing weekly website visits: [120, 150, 135, 180, 210, 195, 240]. Label the x-axis with Day 1 to Day 7 and add a title.

```
import matplotlib.pyplot as plt
visits = [120, 150, 135, 180, 210, 195, 240]
days = ["Day 1", "Day 2", "Day 3", "Day 4", "Day 5", "Day 6", "Day 7"]
plt.plot(days, visits, marker='o')
plt.xlabel("Days")
plt.ylabel("Website Visits")
plt.title("Weekly Website Visits")
plt.show()
```

2. Create a bar chart comparing marks in Python, NumPy, Pandas, and Matplotlib: [82, 90, 76, 88]. Add axis labels and display the chart.

```
import matplotlib.pyplot as plt

subjects = ["Python", "NumPy", "Pandas", "Matplotlib"]
marks = [82, 90, 76, 88]
plt.bar(subjects, marks)

plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("Marks Comparison")
plt.show()
```

3. Plot two lines on the same chart for Product A sales [10, 14, 13, 18, 22] and Product B sales [8, 12, 15, 16, 20]. Add a legend to distinguish them.

```
import matplotlib.pyplot as plt

product_a = [10, 14, 13, 18, 22]
product_b = [8, 12, 15, 16, 20]
```

```
days = [1, 2, 3, 4, 5]

plt.plot(days, product_a, marker="o", label="Product A")
plt.plot(days, product_b, marker="o", label="Product B")

plt.xlabel("Days")
plt.ylabel("Sales")
plt.title("Product A vs Product B Sales")
plt.legend()
plt.show()
```

Part B – Histogram and Scatter Plot

4. Create a histogram from exam scores [45, 52, 58, 60, 61, 67, 72, 75, 81, 84, 89, 92, 96]. Set a suitable number of bins and title the chart.

```
import matplotlib.pyplot as plt

scores = [45, 52, 58, 60, 61, 67, 72, 75, 81, 84, 89, 92, 96]

plt.hist(scores, bins=5)

plt.xlabel("Exam Scores")
plt.ylabel("Frequency")
plt.title("Distribution of Exam Scores")

plt.show()
```

5. Create a scatter plot for study hours [1, 2, 3, 4, 5, 6] and marks [48, 55, 65, 72, 83, 91]. Label both axes and give the plot a title.

```
import matplotlib.pyplot as plt

study_hours = [1, 2, 3, 4, 5, 6]
marks = [48, 55, 65, 72, 83, 91]
plt.scatter(study_hours, marks)
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```

6. Observe the scatter plot and write one or two sentences about the relationship between study hours and marks.
   Ans: As study hours increase, marks also increase. This shows that students who study more tend to get higher marks.

Part C – Formatting and Saving

7. Add a legend to a chart that contains more than one data series.

```python
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y1 = [10, 20, 30, 40, 50]
y2 = [15, 25, 35, 45, 55]
plt.plot(x, y1, label="Series 1")
plt.plot(x, y2, label="Series 2")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Multiple Data Series")
plt.legend()
plt.show()
```

8. Add a grid to one chart and change the marker style on the line chart.

```python
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 30, 40, 50]
plt.plot(x, y, marker="o")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Line Chart with Markers")
plt.grid(True)
plt.show()
```

9. Save one of your plots as student_performance.png using savefig().

```python
import matplotlib.pyplot as plt

study_hours = [1, 2, 3, 4, 5, 6]
marks = [48, 55, 65, 72, 83, 91]
plt.plot(study_hours, marks, marker="o")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Student Performance")
plt.grid(True)
plt.savefig("student_performance.png")
plt.show()
```

10. Write briefly when you would choose a line chart, bar chart, histogram, or scatter plot.

- **Line chart:** I use it to show changes over time, like monthly sales.
- **Bar chart:** I use it to compare values between categories, like marks in different subjects.
- **Histogram:** I use it to understand the distribution of data, like students' exam scores.
- **Scatter plot:** I use it to see the relationship between two things, like study hours and marks.

3. Submission

Submit matplotlib_assignment_YourName.py and the saved image student_performance.png.

4. Submission Checklist

☐ I completed the coding tasks.

☐ I ran the program and checked the output.

☐ I used clear variable names and comments where useful.

☐ I can explain the main concepts.

Key Principle: Practice the concept, run the code, and explain the result.